Program 7: IP Addressing

AIM: Write a program to implement Classful addressing and Classless addressing

Classful Addressing : In classful addressing, IPv4 addresses are divided into Class A, B, C, D, and E based on the first octet.

Class A: 1–126
Class B: 128–191
Class C: 192–223
Class D: 224–239 — multicast
Class E: 240–255 — experimental

Program:

```
#include <stdio.h>
#include <stdio.h>
int main()
{
   int a, b, c, d;

   printf("Enter IPv4 address: ");
   scanf("%d.%d.%d.%d", &a, &b, &c, &d);

   // Check whether the IP address is valid
   if (a < 0 || a > 255 ||
      b < 0 || b > 255 ||
      c < 0 || c > 255 ||
      d < 0 || d > 255) {
      printf("Invalid IP address\n");
      return 0;
   }
   // Find the class using the first octet
   if (a >= 1 && a <= 126)
      printf("Class A\n");
   else if (a >= 128 && a <= 191)
      printf("Class B\n");
   else if (a >= 192 && a <= 223)
      printf("Class C\n");
   else if (a >= 224 && a <= 239)
      printf("Class D\n");
   else if (a >= 240 && a <= 255)
      printf("Class E\n");
   else
      printf("Reserved / Invalid for classful addressing\n");

   return 0;
}
```

OUTPUT:

```
Enter IPv4 address: 192.168.1.10
Class C
```

Classless addressing:
Classless addressing uses a prefix length, such as /24, /16, or /20, instead of fixed A/B/C classes.
For example:

192.168.1.0/24

Here /24 means the first 24 bits are the network portion.

C program to calculate the subnet mask from a CIDR prefix:

```
#include <stdio.h>
#include <stdint.h>

int main() {
   unsigned int a, b, c, d;
   int n;
   uint32_t ip, mask, network, broadcast;

   printf("Enter IP address with prefix (example: 192.168.1.10/24): ");
   scanf("%u.%u.%u.%u/%d", &a, &b, &c, &d, &n);

   // Validate input
   if (a > 255 || b > 255 || c > 255 || d > 255 || n < 0 || n > 32) {
      printf("Invalid IP address or prefix length.\n");
      return 1;
   }

   // Convert IP address to 32-bit value
   ip = (a << 24) | (b << 16) | (c << 8) | d;

   // Create subnet mask
   if (n == 0)
      mask = 0;
   else
      mask = 0xFFFFFFFF << (32 - n);

   // Calculate network and broadcast addresses
   network = ip & mask;
   broadcast = network | ~mask;

   printf("\nCIDR Prefix: /%d", n);

   printf("\nSubnet Mask: %u.%u.%u.%u",
         (mask >> 24) & 255,
         (mask >> 16) & 255,
         (mask >> 8) & 255,
         mask & 255);

   printf("\nNetwork Address: %u.%u.%u.%u",
         (network >> 24) & 255,
         (network >> 16) & 255,
         (network >> 8) & 255,
```

```
        network & 255);

    printf("\nBroadcast Address: %u.%u.%u.%u",
        (broadcast >> 24) & 255,
        (broadcast >> 16) & 255,
        (broadcast >> 8) & 255,
        broadcast & 255);

    if (n == 32)
        printf("\nUsable Hosts: 1\n");
    else if (n == 31)
        printf("\nUsable Hosts: 2\n");
    else
        printf("\nUsable Hosts: %u\n", (1U << (32 - n)) - 2);

    return 0;
}
```

Input: Enter IP address with prefix (example: 192.168.1.10/24): 192.168.1.10/24
Output: CIDR Prefix: /24
Subnet Mask: 255.255.255.0
Network Address: 192.168.1.0
Broadcast Address: 192.168.1.255
Usable Hosts: 254

What is Classful addressing